

Daniel Oppenheimer

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Education

- **University of Virginia** **Aug 2023 – May 2027**
- BS in Computer Science, Data Science Minor, Engineering Business Minor
- Cumulative GPA: 3.78
- Relevant Coursework: Cloud Computing, Parallel Computing, Machine Learning, Artificial Intelligence, Computer Networks, Natural Language Processing, Software Development Essentials, Software Engineering, Data Structures and Algorithms I & II, Computer Systems and Organization I & II, Discrete Mathematics I & II, New Product Development, Data Science Systems
- Activities and Societies: Trigon Engineering Society @ UVA, Sigma Alpha Mu – Special Events Chair, 2025 Deloitte National Undergraduate Case Competition, Intramural Sports Supervisor

Experience

Comcast, Software Engineer Intern

June 2026 – August 2026

- Streamlined Xfinity's CI/CD migration to GitHub Actions and fixed flaky tests, reducing development build failures by 20% and eliminating hour-long release delays for a customer-facing product serving 30M users.
- Improved technical infrastructure for a team of ~50 by reducing API log clutter with reusable batching system and built tools to update stale development routes/allow for production payment flow testing.
- Architected new Wi-Fi troubleshooting menu for Xfinity app to drastically reduce frivolous customer support contacts and customer churn rate – earned commendation from CTO.

Booz Allen Hamilton, AI Software Engineer Intern

June 2025 – August 2025

- Engineered and benchmarked three distinct Retrieval-Augmented Generation (RAG) frameworks for a NOAA search tool, collaborating with their developers to enhance retrieval from a complex data storage system.
- Implemented a novel hybrid search system integrating BM25 keyword ranking with semantic search, leveraging a Reciprocal Rank Fusion pipeline and a genetic algorithm to optimize search result relevancy.
- Developed a full-stack solution with a Flask frontend, connecting to a backend on AWS EC2. Used REST API strategies to connect to NOAA data, a Redis cache, and the Gemma3 GenAI model.
- Developed CoT system to auto-upload new data formats in a metadata-rich format.

Zbooni, Software Development Consultant

Dec 2025 – Jan 2026

- Engineered a centralized fraud detection dashboard to automate risk assessment for the compliance team, replacing manual oversight with robust security rules, saving KYC team members 3 hours per workday.
- Architected an automated email verification workflow using Trengo for suspicious transactions, streamlining customer confirmation responses and significantly reducing the platform's exposure to fraudulent activity.
- Prototyped Random Forest and XGBoost models for risk data collected through platform in 6 months.

Glass House Recycling LLC, Co-founder

Feb 2020 – August 2025

- Founded curbside glass-pickup business at age 14 to meet demand after Falls Church City stopped providing municipal glass recycling service.
- Developed in-house invoicing system with a Node.js express web application.
- Leveraged digital advertising and print/radio media (NBC, WAMU, NowThis) to grow customer base to over 160 residential households, online stories about business reached 550k+ views, with 5k+ reactions.
- \$10k+ exit, selling company and converting over 80% of customers to the new platform.

Projects

Bracket Hub

Apr 2024 – Nov 2025

- Built social web application to bring a March Madness pointing system to user-made tournament brackets.
- Used React.js interfaced with Firebase authentication and Firestore database for backend and data storage.

Citi-Defense

May 2026 – Present

- Engineered 1-bit tower-defense game for an 8 MHz Macintosh Plus with only 1 MB of RAM.
- Built memory-efficient sprite compositor and asset pipeline using fixed-point, integer-only systems.

Bide

June 2026 – Present

- Built full-stack iOS application to surface better ETA and carpool information for friends/family meetups
- Integrated PostgreSQL, Apple MapKit, In-App purchases, and iMessage tiles for an ideal user experience

Spyfall

Jan 2020 – Apr 2020

- Designed minimalist reimagination of the “Spyfall” game using Node.js + Socket.io for live multiplayer

Skills

- Python, JavaScript (React, Node.js, Typescript), Java, HTML/CSS, SQL
- Git, PyTorch, RAG, Linux, GitHub Actions, Concourse, Docker, Ollama, Firebase, Flask, Django, REST API, MATLAB, Agile Method, Jupyter Notebook, Blender

Awards: Eagle Scout